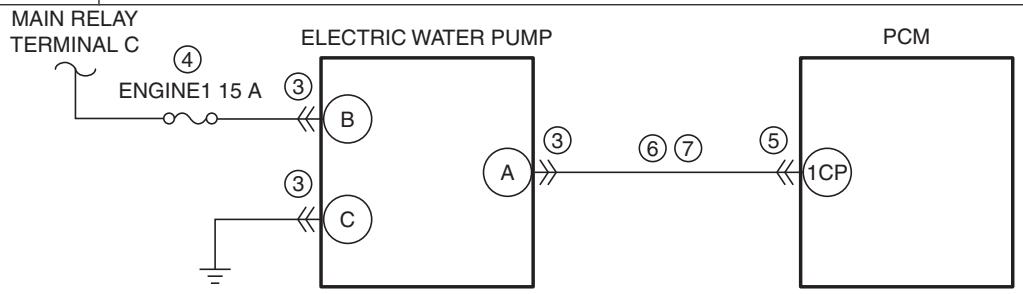
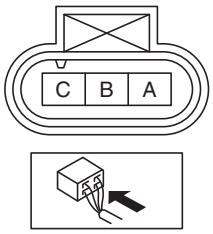
DTC P0A06:00 [SKYACTIV-D 1.5]

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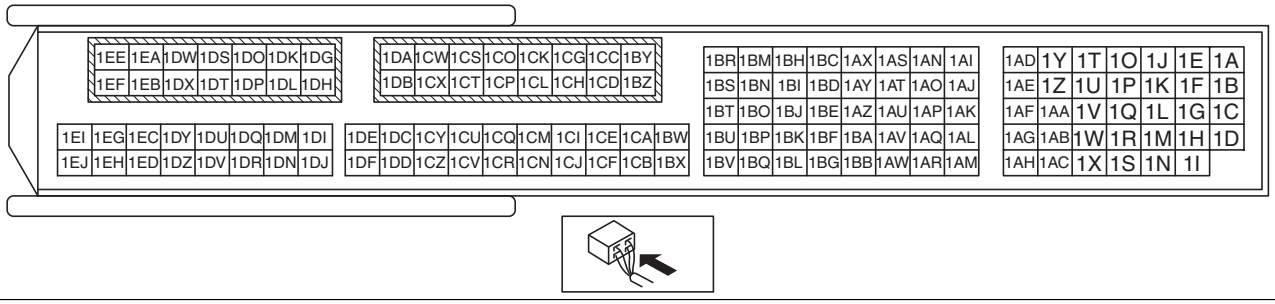
DTC P0A06:00	Electric water pump control circuit low input
DETECTION CONDITION	• When the following conditions are met, the accumulated time under a condition in which the electric water pump control circuit voltage is **less than 35%** of the battery voltage is **5 s or more.** **MONITORING CONDITIONS** — All of the following conditions is met for 1min: — Duty value of the electric water pump: **80 % or more** • Battery voltage: **11—20 V** • Duty value of the electric water pump: **30—100 %** **Diagnostic support note** • This is a continuous monitor (other). • The check engine light does not illuminates. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Electric water pump connector or terminals malfunction • Short to ground or open circuit in electric water pump power supply circuit — Short to ground in wiring harness between the main relay terminal C and electric water pump terminal B — ENGINE1 15 A fuse malfunction — Open circuit in wiring harness between the main relay terminal C and electric water pump terminal B • PCM connector or terminals malfunction • Short to ground in wiring harness between the electric water pump terminal A and PCM terminal 1CP • Open circuit in wiring harness between the electric water pump terminal A and PCM terminal 1CP • Electric water pump malfunction • PCM malfunction



ELECTRIC WATER PUMP
WIRING HARNESS-SIDE CONNECTOR



PCM
WIRING HARNESS-SIDE CONNECTOR



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT ELECTRIC WATER PUMP CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the electric water pump connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.
4	**INSPECT ELECTRIC WATER PUMP POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the electric water pump connector is disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the electric water pump terminal B (wiring harness-side). • Is the voltage B+?	Yes	Go to the next step.
		No	Inspect the ENGINE1 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and electric water pump terminal B. If there is a common connector: • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. If there is no common connector: • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and electric water pump terminal B. If there is a common connector: • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. If there is no common connector: • Repair or replace the wiring harness which has an open circuit. Go to Step 8.
5	**INSPECT PCM CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 8.
		No	Go to the next step.

STEP	INSPECTION		ACTION
6	**INSPECT ELECTRIC WATER PUMP CONTROL CIRCUIT FOR SHORT TO GROUND** • Verify that the electric water pump and PCM connectors are disconnected. • Inspect for continuity between the electric water pump terminal A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between electric water pump terminal A and PCM terminal 1CP. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 8.
		No	Go to the next step.
7	**INSPECT ELECTRIC WATER PUMP CONTROL CIRCUIT FOR OPEN CIRCUIT** • Verify that the electric water pump and PCM connectors are disconnected. • Inspect for continuity between electric water pump terminal A (wiring harness-side) and PCM terminal 1CP(wiring harness-side). • Is there continuity?	Yes	Replace the electric water pump, then go to the next step. (See ELECTRIC WATER PUMP REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Refer to the wiring diagram and verify whether or not there is a common connector between electric water pump terminal A and PCM terminal 1CP. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
8	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOEO self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
9	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.